

Module 06: Learning in Robotics — Bio-Inspired Adaptation

Learning Objectives

1. Explain how **biological learning mechanisms** (synaptic plasticity, neuromodulation, developmental learning) inspire robotic learning architectures in Active Inference.
2. Analyze the role of **Dirichlet parameter learning** and **structure learning** (Bayesian Model Reduction) in robotic model adaptation.
3. Apply bio-inspired learning to achieve **sample-efficient adaptation** in robots that learn from limited experience.

Introduction

Biological organisms learn continuously and efficiently — a child learns to walk within months, acquiring motor competence from a remarkably small number of falls. This efficiency stands in stark contrast to deep reinforcement learning, which may require millions of simulated episodes to learn the same task. Bio-inspired robotic learning aims to close this gap by replicating the mechanisms that make biological learning so efficient: structured priors, curiosity-driven exploration, and hierarchical model updating.

Key Concepts

1. Parameter Learning: Updating the Generative Model

The most basic form of robotic learning is **parameter learning** — updating the A and B matrices of the generative model based on experience:

- **A matrix learning:** The robot observes sensor readings in known states, updating its observation model. Example: a mobile robot learns that its sonar returns different echo profiles in carpeted vs. tiled rooms.

- **B matrix learning:** The robot observes state transitions given its actions, updating its transition model. Example: a manipulation robot learns that grasping a wet object requires different predicted forces than grasping a dry one.

In Active Inference, parameter learning uses **Dirichlet distributions** — the concentration parameters grow with experience, making the model increasingly confident. Early experiences have large effects (high learning rate); later experiences refine precision (low learning rate). This mirrors biological synaptic plasticity, where initial learning is rapid and later experience fine-tunes existing connections.

2. Structure Learning: Bayesian Model Reduction

Bayesian Model Reduction (BMR) enables a robot to simplify its generative model after accumulating experience — pruning unnecessary states, actions, or observation channels that are not supported by data:

- A robot that initially represents 20 possible obstacle types discovers that only 5 types are encountered in its environment → BMR prunes the unused states
- A multi-sensor robot discovers that one of its cameras is consistently uninformative → BMR reduces its precision to near-zero, effectively ignoring it

BMR is the computational analog of **synaptic pruning** in brain development — the brain initially overproduces synapses and then eliminates those not reinforced by experience.

3. Curiosity-Driven Exploration

Bio-inspired robots exploit **epistemic value** in their Expected Free Energy to drive curiosity — actively seeking out experiences that maximize information gain:

- A robot exploring a new building preferentially visits rooms it hasn't seen before (high epistemic value) rather than revisiting known rooms (low epistemic value)
- The balance between curiosity (epistemic) and task completion (pragmatic) is automatically regulated by the EFE — in early exploration, epistemic value dominates; as the model matures, pragmatic value dominates

This mirrors infant exploratory behavior: babies are compulsive experimenters, dropping objects, mouthing textures, and poking surfaces — all high-epistemic-value actions that rapidly build their generative model.

4. Developmental Learning

Bio-inspired developmental learning structures the learning process across stages, inspired by human cognitive development:

- **Stage 1 (Sensorimotor):** Learning basic sensor-motor contingencies — "when I move my arm, the visual field changes"
- **Stage 2 (Object permanence):** Learning that objects persist when unobserved — building a state space that includes hidden states
- **Stage 3 (Causal reasoning):** Learning temporal regularities — "pushing this button unlocks that door"
- **Stage 4 (Social learning):** Learning from observing other agents — imitation as generative model transfer

Applications

- **One-shot grasp learning:** A robot that has deeply learned general grasping priors (from bio-inspired developmental training) can adapt to a novel object in a single trial — the deep generative model provides strong structural priors, and a single experience updates the object-specific parameters.
- **Lifelong mobile robot deployment:** A service robot deployed in a hospital learns room layouts, staff schedules, and equipment locations over months of operation, using BMR to prune outdated information (a room was renovated, a staff member retired) and curiosity to explore new areas.

Conclusion

Bio-inspired learning gives robots the sample efficiency and adaptiveness of biological organisms. Parameter learning, structure learning, curiosity-driven exploration, and developmental staging all implement Active Inference learning mechanisms. The next module examines how bio-inspired communication enables multi-robot cooperation.